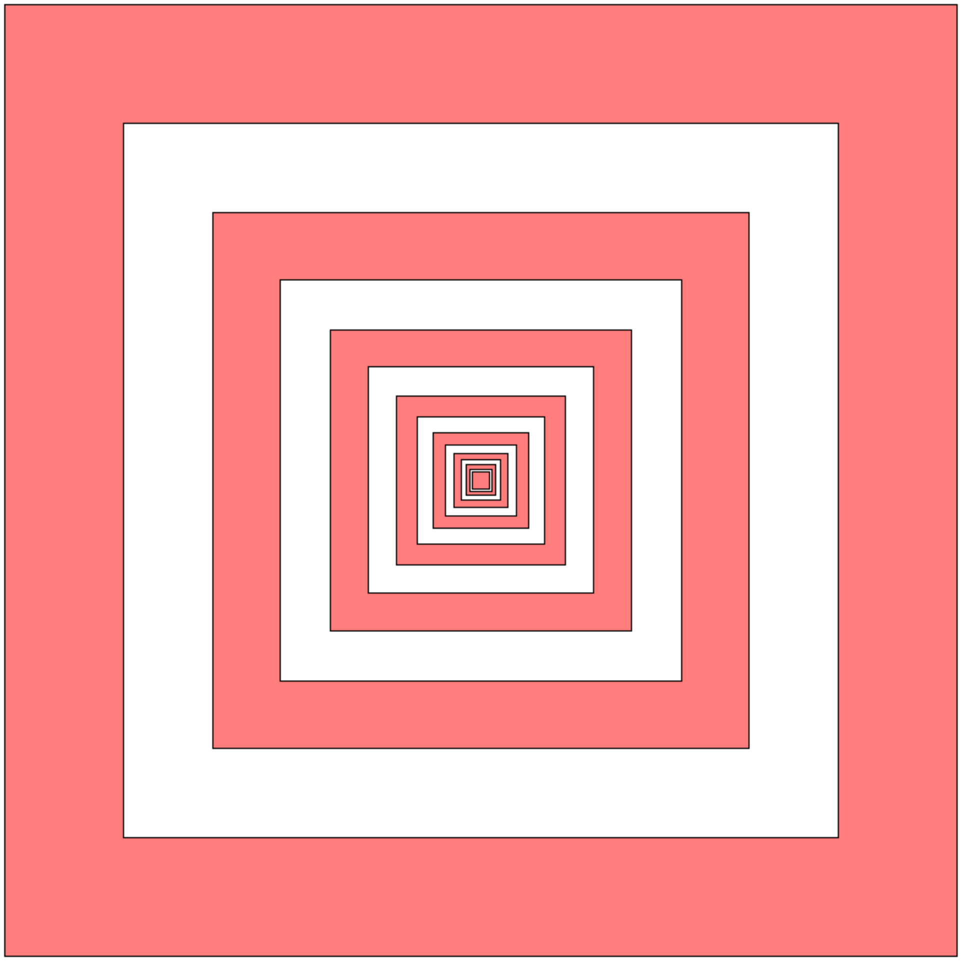
2025 AMC 10A Problem 13 - Solution

Review the full statement and step-by-step solution for 2025 AMC 10A Problem 13. Great practice for AMC 10, AMC 12, AIME, and other math contests

2025 AMC 10A Problem 13

Problem:
In the figure below, the outside square contains infinitely many squares, each of them with the same center and sides parallel to the outside square. The ratio of the side length of a square to the side length of the next inner square is k, where $0 < k < 1$. The spaces between squares are alternately shaded, as shown in the figure (which is not necessarily drawn to scale).



The area of the shaded portion of the figure is 64% of the area of the original square. What is $k?$

Answer Choices:

A. $\dfrac{3}{5}$

B. $\dfrac{16}{25}$

C. $\dfrac{2}{3}$

D. $\dfrac{3}{4}$

E. $\dfrac{4}{5}$

Solution:

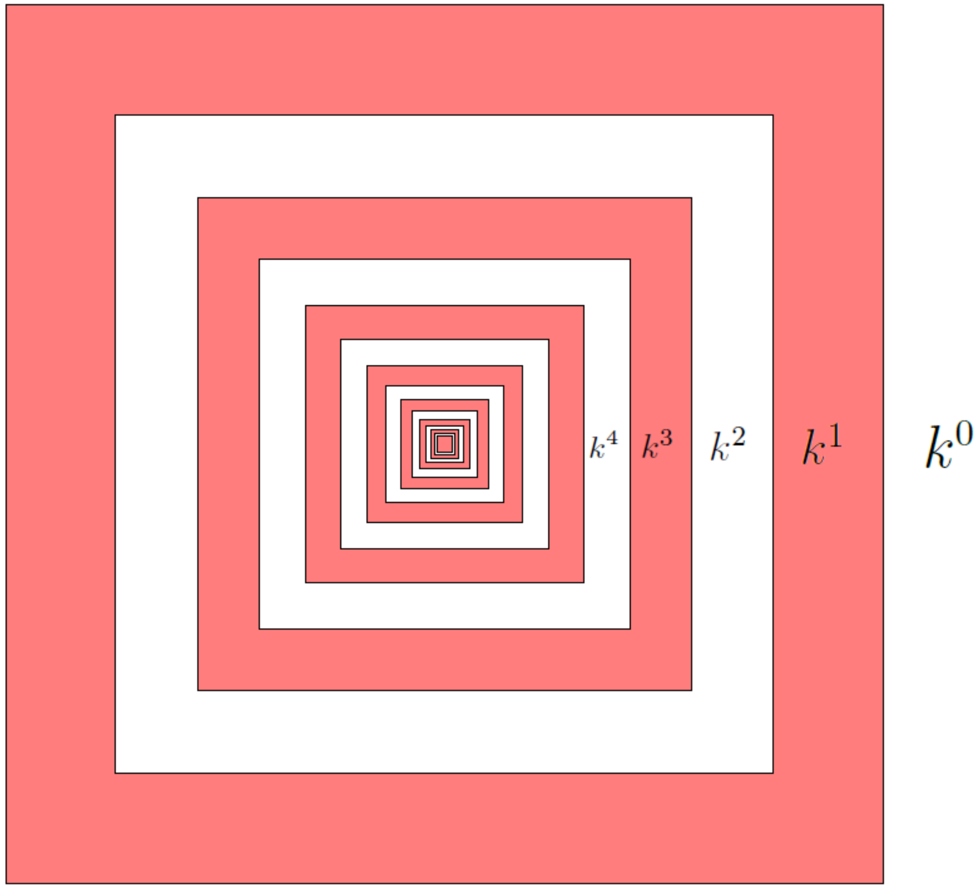
Let the side length of the largest (outer) square be 1. Each successive inner square has side length multiplied by a factor of k $(0 < k < 1)$, so the side lengths are

$$1,\ k,\ k^2,\ k^3,\ k^4,\ \ldots$$

and the corresponding areas are

$$1,\ k^2,\ k^4,\ k^6,\ k^8,\ \ldots$$

The shaded area, which are the rings, are just the difference between consecutive squares. For example, the first ring is $1 - k^2$, second $k^4 - k^6$, and so on.



Thus the total shaded area (as a fraction of the area of the largest square) is

$$1 - k^2 + k^4 - k^6 + k^8 - k^{10} + \dots$$

This is an infinite geometric series with first term 1 and common ratio $-k^2$, so

$$1 - k^2 + k^4 - k^6 + k^8 - k^{10} + \cdots = \frac{1}{1+k^2}.$$

We are told that this shaded portion is 64% of the area of the original square, so

$$
\begin{aligned}
&1-k^2+k^4-k^6+k^8-k^{10}+\cdots=\frac{1}{1+k^2}\\
\Rightarrow\quad &\frac{1}{1+k^2}=\frac{64}{100}=\frac{16}{25}\\
\Rightarrow\quad &16k^2=9\Rightarrow k=\frac{3}{4}.
\end{aligned}
$$

Therefore, k = $\boxed{\textbf{(D) } \frac{3}{4}}$